FILTER DESIGN TEST CASE

Notch Filter - Band 1 (2100 MHz) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Notch Filter filter targeting Band 1 (2100 MHz) for Public Safety Band Filter. The filter is designed on AlN on SiC substrate with a target center frequency of 5151.1 MHz and bandwidth of 176.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Notch Filter
- Frequency Band: Band 1 (2100 MHz)
- Application: Public Safety Band Filter
- Substrate: AlN on SiC
- Package: Fan-Out WLP
- Design Tool: Matlab RF Toolbox R2024b
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the Notch Filter design targeting Band 1 (2100 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 5151.1 MHz
Bandwidth (3 dB): 176.7 MHz
Insertion Loss Target: <= 2.2 dB
Return Loss Target: >= 22.0 dB
Out-of-Band Rejection: >= 44.0 dB
Isolation (Duplexer): >= 20.0 dB
Q Factor: 8292
Temperature Coefficient:-25.2 ppm/°C
Die Size: 2.0 x 2.0 mm
Wafer Size: 8-inch
Number of Resonators: 10
Electrode Material: W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Matlab RF Toolbox R2024b and load the Notch Filter template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 5151.1 MHz, BW 176.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on SiC).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15453 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.2 dB
- Return loss in passband shall be >= 22.0 dB
- Out-of-band rejection at +/- 265 MHz offset >= 44.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 16.2 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 90%

6. TEST RESULTS

Measurement Date: 2025-02-20
Devices Tested: 13
Insertion Loss (meas): 1.86 dB
Return Loss (meas): 24.5 dB
Rejection (meas): 47.5 dB
Isolation (meas): 24.7 dB
Q Factor (meas): 8292
Group Delay Variation: 2.7 ns
Power Handling: +28 dBm
Die Yield: 80.3%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

a) The Notch Filter design demonstrated below-target performance for Band 1 (2100 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

1. Review near-band rejection margin for worst-case temperature.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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Approver: Dr. Karen Mitchell Date: 2025-02-27

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